

CMPT 280

Topic 2: Regression Testing

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References

- Textbook, Chapter 2

Regression Testing

- To write regression tests for our Java data structure classes, we will use the `main()` method for this class.
- This does not mean that this `main()` method is an entry point a the program that the data structure is used in. It is only a test of the class.
- A regression test in a `main()` program is only run when you want to repeat the tests for the class, perhaps after a modification, bugfix, or feature addition.
- A regression test provides quick way to verify the correctness of the class before using it in part of a larger program.

Regression Testing

```
1 public class LinkedList<I> implements ListADT<I> {  
2     protected ListNode<I> head;  
3     protected int numEl;  
4  
5     // ... Other methods and constructors ...  
6  
7     public static void main(String args[]) {  
8         // Regression test goes here.  
9     }  
10 }
```

Exercise 1

Implement some tests for the `ArrayList` class:

- Test that `isEmpty()` returns correct value when the list is actually empty.
- Test `firstItem()` when the list is empty.
- Test `deleteFirst()` when the list is empty.
- Test `insertFirst()` when the list is empty.

(This is not an exhaustive list of test cases)

Exercise 2

- Generate and implement the test cases from Exercise 1 in the `ArrayedList` class.
- Observations:
 - Since the interfaces are the same, we should be able to re-use most of our testing code from `LinkedList`.
 - There will be some differences testing `insertFirst()` and `isFull()`. Why???

Exercise 3

Implement some tests for the `ArrayList` class:

- Write a test for `isFull()` in `ArrayList` when the list is full.
- Write a test for `insertFirst()` in `ArrayList` when the list is full.

What's next?

- Timing analysis of algorithms. We need to be able to compare the efficiency of operations on different data structures and different implementations of the same data structure.
- **Next class reading:** Chapter 3 of textbook.